



Application:

I deal for Hospitals and laboratuvar etc...

Feature:

This LED panel with an antibacterial and air-purifying effect is available in a size of 59.5 x 59.5 cm This is done by circulating air in the room . Many types of bacteria , viruses or fungi to be reduced. For example, dangerous hospital germs (MRSA) , H1N1 or SARS germs.



SPECIFICATIONS

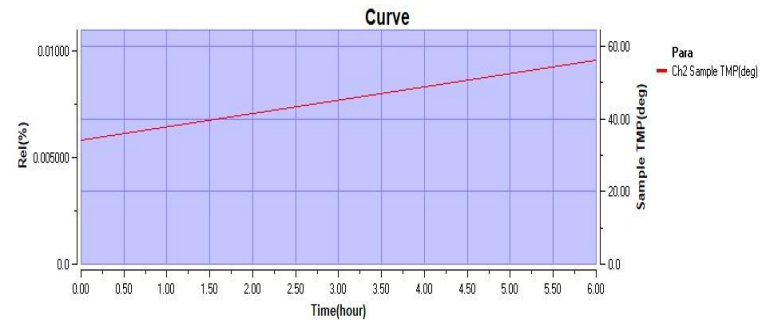
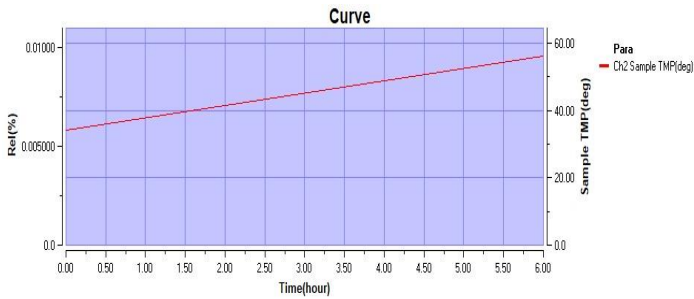
ITEM	DESCRIPTION
Input Voltage	220 - 240 V A C
Frequency	50 / 60 H z
Power factor	>98%
CCT	2700K, 3000K, 4000K, 5000k, 6000K
CRI	>80%
LED Efficiency	120-130Lm/W
Power supply	NASSLI
LED Type	Osram/Epistar
Housing	Steel sheet with electrostatic powder coating
Beam Angle	115°
Life Time	>50,000 Hour
Operating Temp	-20°C to +55°C
Storage Temp	-40°C to +65°C
Certifications	CE, SELV and ISO
Mounting	Recessed / Suspended / Surface
IP Rating	IP65
Dimmable	On Request
Emergency	On Request
Warranty	5 Years

Order code	Dimensions(mm)	Power(Watt)	Lumen
NSL-PL-ANB-6060-42W-XXK-SMD-NS-IP65	595x595x90	42	3760

Test Report (ISTMT)

SP-77092-2-WAS

PASS



SN	Time	Aging(hour)	Ch1 Voltage(V)	Room TMP(deg)	LIGHT BAR(TOSPO)	LIGHT BAR(TOSPO)	TC	POWER	DRIVER TCI (122213)	IF	TC
1	9/29/2022 8:10	0	220.5	39.7	43.4	41.4	85	72	34.3	550	85
2	9/29/2022 14:10	6	220.43	42.6	53.5	49.4			56.1		



NEXT GENERATION OF PERFORMANCE PLASTIC – ANTIMICROBIAL ACRYLIC SHEET

TRANSPARENT SHEET FORMULATED WITH A SILVER ION ANTIMICROBIAL AGENT THAT PROTECTS THE SURFACE FROM MICROORGANISM GROWTH

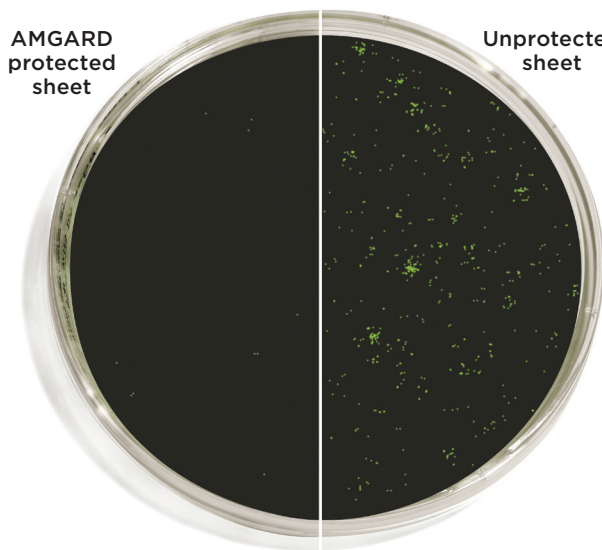
Safety Shields can help prevent the spread of airborne illnesses by creating a physical barrier between people, but they also present a surface for microorganisms to grow. Silver ion antimicrobial technology protects AMGARD surfaces from degradation by preventing the spread and growth of microorganisms. The technology is built into AMGARD during the manufacturing process and remains active for the life of the product.

Certified laboratories have independently tested AMGARD to ISO 22196 “Measurement of Antimicrobial Activity on Plastics / Non-Porous Surfaces” – the industry standard for measuring Antimicrobial activity on Plastics and confirmed a reduction in the growth of microorganisms on the sheet surface.

AMGARD is available as an OPTIX® Acrylic sheet option to meet a range of applications

- » Protects surface against the growth of microorganisms such as bacteria, mold and mildew that cause stains and odors
- » Provides additional surface protection between cleanings
- » Complies with applicable EPA requirements as a Treated Article
- » Stocked in 48”x 96” sheet of 0.220” thickness
- » Tested to ISO 22196 “Measurement of Antimicrobial Activity on Plastics / Non – Porous Surfaces”

AMGARD
protected
sheet



Unprotected
sheet

Representative 24-hour antimicrobial effect. AMGARD is formulated with silver ion antimicrobial technology to protect the sheet’s surface. It is not designed to prevent the transmission of any disease or infectious agent.

Product Comparison

	Glass	Regular Sheet	AMGARD Sheet
Prevents Microbe Growth	No	No	Yes
Shatter Resistant	No	Yes	Yes
Lightweight	No	Yes	Yes



DISINFECTING ACRYLIC

When disinfecting AMGARD Safety Shields, proper procedures should be taken. To minimize the risk of damage, use only compatible household cleaners as outlined below.

For more details please contact the Plaskolite Technical Service Group at **800.848.9124** with any questions.

Disinfectants for Plaskolite Products

Product/ Brand	Manufacturer	Products	Compatibility**
AMGARD Acrylic	Generic	Hydrogen peroxide 3%-5%	Compatible
AMGARD Acrylic	Generic	Bleach - household laundry grade	Compatible
AMGARD Acrylic	Generic	Quaternary ammonium compounds @ use dilution	Compatible
AMGARD Acrylic	Generic	Isopropyl alcohol - diluted with water to 30% strength	Compatible
AMGARD Acrylic	Generic	Warm dish washing soap solution	Compatible
AMGARD Acrylic	Generic	Ethyl alcohol - pharmacy grade	Not Compatible

SPECIAL NOTES: Cleaning products were tested on flat uncoated AMGARD Acrylic sheet products. Do not use incompatible cleaners or disinfectants on sheet products. Generally speaking, soap and water or hydrogen peroxide disinfectants work best on these products.

**Because manufacturers sometimes change their formulas without notification compatibility in perpetuity can not be presumed.

These suggestions and data are based on information we believe to be reliable. They are offered in good faith, but without guarantee, as conditions and methods of use are beyond our control. We recommend that the prospective user determine the suitability of our materials and suggestions before adopting them on a commercial scale.

PLASKOLITE

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CERTIFICATE OF ANTIBACTERIAL ANALYSIS

CERTIFICATE NO.	BC098/2020	DATE RECEIVED	09.07.2020
CUSTOMER	AKZONOBEL	DATE ANALYSED	18.08.2020
CUSTOMER REF.	200/553	DATE REPORTED	21.08.2020
MANUFACTURER	ITALY		
SAMPLE COMMENT	LWR POW324883/2		
UNITS OF RESULTS	Colony Forming Units	NO. OF PAGES	1 of 1

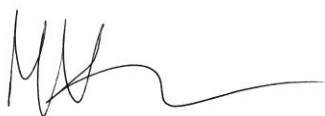
METHOD OF ANALYSIS: DETERMINATION OF ANTIBACTERIAL ACTIVITY USING ISO 22196:2011

SAMPLE	TEST ORGANISM	CONTACT TIME		REDUCTION (INITIAL)	
		0 HRS	24 HRS	Log ₁₀	%
INTERPON 610 (LOW-E) AM	MRSA	3.31E+04	≤100	≥2.52	≥99.70%
INTERPON 610 (LOW-E) AM	<i>E. coli</i>	6.73E+04	≤100	≥2.83	≥99.85%

The above data describe the difference in the population sizes of the test organisms, relative to the initial (0 hours) population, following contact with the surface of the samples detailed in this CoA for 24 hours at 35°C under a RH of >90%. These conditions are those specified by the ISO 22196: 2011 method of analysis.

Comment: The sample INTERPON 610 (LOW-E) AM has achieved the BioCote minimum antibacterial performance requirement of 95% "Reduction against the initial for *E. coli* and MRSA" according to ISO 22196: 2011 analysis.

FOR BIOCOTE LTD



Technical Manager

Megan Vaughan

Product Data Sheet

AkzoNobel Powder Coatings

Interpon 610 AM

MA260I INT 610 AM WHITE SN20

Product Description

Interpon 610 AM is a series of polyester based powder coatings, formulated without the use of TGIC, designed for the exterior environment, offering excellent light and weather resistance from a single coat finish on a variety of substrates in combination with specific antimicrobial activity.

Interpon 610 AM powders are available in a wide range of colors in gloss, reduced gloss, textured and other special finishes or can be custom matched to the user's requirements.

Powder Properties

Chemical type	Polyester TGIC Free
Density (g/cm³)	1.20-1.90 g/cm ³ ±0.03 depending on color and effect
Application	Suitable for electrostatic spray
Storage	Under dry, cool (≤ 30°C) conditions (open boxes must be resealed)
Shelf life	24 months below 30°C 12 months below 35°C
Curing schedule	20-40 minutes at 170°C 10-20 minutes at 180°C 8-16 minutes at 200°C

Test Conditions

The results shown below are based on mechanical and chemical tests which (unless otherwise indicated) have been carried out under laboratory conditions and are given for guidance only. Actual product performance will depend upon the circumstances under which the product is used.

Substrate	Gold Seal polished 0.5mm steel
Pretreatment	Gold Seal lightweight Zinc Phosphate
Film Thickness	80 microns
Curing Schedule (Object Temperature)	10 minutes at 200°C

Mechanical Tests

Adhesion	ISO 2409 (2mm crosshatch)	Gt 0
Flexibility	ISO 1519	Pass 5mm
Erichsen Cupping	ISO 1520	≥ 5mm
Impact	ISO 6272	> 30 kg·cm direct

Chemical and durability tests

Whilst maintaining the general protective and anti-corrosive properties of powder coatings, aluminum and copper/bronze metallic finishes, when submitted to the listed tests, may rapidly show a loss of metallic aspect.

The results shown are based on tests which (unless otherwise indicated) have been carried out under laboratory conditions and are given for advice only, actual performance depends upon the circumstances under which the product is used.

Salt Spray (250 hours)	ISO 9227	no corrosion creep more than 3 mm from scribe
Exterior Durability	Suitable for outdoor exposure.	
Chemical Resistance	Generally good resistance to acid, alkalis and oil at room temperatures	

For application on Buildings we recommend using our *Interpon D* series.

Pretreatment

Aluminium, steel or Zintec surfaces to be coated must be clean and free from grease. Iron phosphate and particularly lightweight zinc phosphating of ferrous metals improves corrosion resistance. Aluminium substrates may require a chromate conversion coating.

Detailed advice should be sought from the pre-treatment supplier.

Application

Interpon 610 powder coatings can be applied by corona electrostatic equipment.

In all application processes the aspect obtained is subject to variation, depending on the method of application (type of gun, nozzle, etc) and the shape/type of component. We recommend that the actual application parameters are adapted and adjusted depending on the type of component and with each powder batch to give a finish in accordance with our color card.

The following procedure is given as a guideline when using these finishes. We recommend the use of flat jet spray nozzles. To ensure powder homogeneity, the complete content of the boxes should be emptied completely into the feed hopper. For manual application it is essential to ensure that an even film thickness is applied, and, in all instances, sinusoidal gun movements should be avoided.

All powders can show small color differences from batch to batch, this is normal and unavoidable. While AkzoNobel take every precaution to minimize visible differences, this cannot be guaranteed. Applicators and fabricators are advised to use a single batch for parts that will be assembled together. Differences are more likely with special effect powders.

Bonded products have better application properties than blended products (more stable) but attention should still be paid to line settings to avoid "marble effect" and changes in aspect after recycling. For more details it is suggested to read the "**Metallic Application Guideline**".

Different substrates (Aluminium, steel, galvanized steel, etc.), use of primer, and big changes in film thickness may give a different aspect. Products with different codes should not be mixed even if same color and gloss.

Recycling	Unused powder can be reclaimed using suitable equipment and recycled through the coating system, but a minimum of 70% virgin powder should be used.
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Post Application

Contact with Chemical Agents

Contact, even for a short duration with certain household products and chemicals, can cause irreversible changes in the gloss and appearance.

We recommend that a test is carried out on a non-visible area before using these types of products on this coating.

Additional Information

Interpon 610 AM in conjunction with BioCote Ltd ® has been tested for antimicrobial efficacy in accordance with ISO 22196: 2011 and exhibited a minimum of 95% and up to 99.99% reduction in the population of *Escherichia coli* and Methicillin Resistant *Staphylococcus aureus* (MRSA).

Testing was carried out by an independent laboratory and is classified as 'microbiological results satisfactory'. BioCote® silver ion technology has been proven effective against the following bacteria in laboratory conditions:

Multi Drug Resistant Bacteria

- ESBL *Escherichia coli*
- CRE *Klebsiella pneumoniae*
- MRSA Methicillin Resistant *Staphylococcus aureus*
- VRE *Vancomycin Resistant Enterococcus*

Bacteria

- *Acinetobacter baumannii*
- *Bacillus subtilis*
- *Campylobacter* spp.
- *Clostridium difficile* (excluding spore form)
- *Escherichia coli* O157
- *Enterobacter aerogenes*
- *Enterococcus faecalis*
- *Legionella* spp.
- *Listeria monocytogenes*
- *Pseudomonas aeruginosa*
- *Salmonella* Enteritidis
- *Salmonella* Typhimurium
- *Shigella* spp.
- *Staphylococcus aureus*
- *Staphylococcus epidermidis*

Interpon 610 AM contains BioCote® silver phosphate glass antimicrobial technology to preserve the coating surface and prevent degradation caused by microbial growth once applied to the intended substrate.

Safety Precautions

This product is intended for use only by professional applicators in industrial environments and should not be used without reference to the relevant health and safety data sheet which Akzo Nobel has provided to its customers.

Disclaimer

IMPORTANT NOTE: The information in this data sheet is not intended to be exhaustive and is based on the present state of our knowledge and on current laws: any person using the product for any purpose other than that specifically recommended in the technical data sheet without first obtaining written confirmation from us as to the suitability of the product for the intended purpose does so at his own risk. It is always the responsibility of the user to take all necessary steps to fulfil the demands set out in the local rules and legislation. Always read the Material Data Sheet and the Technical Data Sheet for this product if available. All advice we give or any statement made about the product by us (whether in this data sheet or otherwise) is correct to the best of our knowledge but we have no control over the quality or the condition of the substrate or the many factors affecting the use and application of the product.

Therefore, unless we specifically agree in writing otherwise, we do not accept any liability whatsoever for the performance of the product or for any loss or damage arising out of the use of the product. All products supplied and technical advices given are subject to our standard terms and conditions of sale. You should request a copy of this document and review it carefully. The information contained in this data sheet is subject to modification from time to time in the light of experience and our policy of continuous development. It is the user's responsibility to verify that this data sheet is current prior to using the product.

Brand names mentioned in this data sheet are trademarks of or are licensed to AkzoNobel.

<http://www.interpon.com/contact-us/>

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Interpon 610 AM - Issue #3

Last Revision Date: 18.05.2020

Author: Senkyp Petr

AMGARD ACRYLIC SHEET

AMGARD

AMGARD clear acrylic sheet is formulated with a silver ion antimicrobial technology that protects the sheet surface from microorganism growth and remains active for the entire life of the product. When used as protective barriers, AMGARD acrylic sheet provides a clear solid partition between individuals. It is easy to fabricate and form for a wide range of interior applications.

- » Inhibits microbe surface degradation
- » Independent laboratory tested
- » Half the weight of glass
- » Does not shatter into sharp-edged pieces like glass
- » Easy to fabricate & clean

Available in various sheet sizes to maximize yield

APPLICATIONS

Protective barriers, office partitions, safety enclosures, equipment housings

AMGARD is formulated with silver ion antimicrobial technology to protect the sheet's surface. It is not designed to prevent the transmission of any disease or infectious agent.

Typical Properties

Property	Test Method	Units	Values
PHYSICAL			
Specific Gravity	ASTM D792	-	1.19
Light Transmission, Clear @ 0.118"	ASTM D1003	%	92
Water Absorption	ASTM D570	%	0.4
MECHANICAL			
Tensile Strength	ASTM D638	psi	11,030
Tensile Modulus of Elasticity	ASTM D638	psi	490,000
Flexural Strength	ASTM D790	psi	17,000
Izod Impact Strength	ASTM D256	Ft-lb./in	0.4
Rockwell Hardness	ASTM D785		M-95
THERMAL			
Maximum Recommended Continuous Service Temperature	-	°F	170-190
Deflection Temperature @ 264 psi (1.8 MPa)	ASTM D648	°F	203
Coefficient of Linear Thermal Expansion	ASTM D696	in/in/°F	3.0 x 10 ⁻⁵
Flammability (Burning Rate)	ASTM D635	In/min	1.0
Flammability	UL 94	-	HB
Smoke Density Rating	ASTM D2843	%	3.4
Self-Ignition Temperature	ASTM D1929	°F	833

AMGARD ACRYLIC SHEET

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